

Photosynthesis & Cellular Respiration

Dr. Amara Chen · Due August 20, 2026 at 11:59 PM · 40 marks total

Instructions: Answer all questions in complete sentences. Show your work for any calculations. You have 50 minutes to complete this assignment.

ASSIGNMENT QUESTIONS

Question 1 (5 marks)

Which of the following is a product of the light-dependent reactions of photosynthesis?

- A. Glucose
- B. ATP and NADPH
- C. Carbon dioxide
- D. Pyruvate

Model Answer / Expected Response:

- B. ATP and NADPH

Question 2 (15 marks)

Explain the relationship between photosynthesis and cellular respiration. Include the chemical equation for each process and describe how the products of one process serve as the reactants of the other.

Marking Guide / Rubric:

Performance Level	Marks	Criteria
Excellent	15	Both equations correct; clearly explains the cyclical relationship with specific reference to O ₂ , CO ₂ , glucose and water.
Good	11	Both equations mostly correct; explains the relationship with minor gaps.
Developing	6	One equation correct or relationship only partially explained.
Beginning	2	Minimal or incorrect understanding demonstrated.

Model Answer / Expected Response:

Photosynthesis: $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{light} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$

Cellular respiration: $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{energy (ATP)}$

The glucose and oxygen produced by photosynthesis are consumed as reactants in cellular respiration, while the carbon dioxide and water produced by respiration are consumed as reactants in photosynthesis — the two processes form a closed cycle of matter and energy transformation.

Question 3 (20 marks)

Compare and contrast aerobic and anaerobic cellular respiration, including the approximate ATP yield of each pathway per glucose molecule.

Marking Guide / Rubric:

Performance Level	Marks	Criteria
Excellent	20	Correctly contrasts both pathways with accurate ATP yields (~36-38 vs ~2) and identifies the role of oxygen.
Satisfactory	12	Identifies the main difference but ATP yields are vague or partially incorrect.

Model Answer / Expected Response:

Aerobic respiration requires oxygen and yields approximately 36–38 ATP per glucose via glycolysis, the Krebs cycle and the electron transport chain. Anaerobic respiration (fermentation) occurs without oxygen and yields only 2 ATP per glucose via glycolysis alone.